

AMMU ELSA PUNNEN

M. Sc Mathematics | University Rank 8 | Data Analyst

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PROFESSIONAL SUMMARY

University Rank Holder (8th) in M.Sc. Mathematics with a rigorous foundation in abstract logic and statistical modeling. Currently advancing proficiency in Python and SQL through the IBM Data Science Professional Certificate to translate complex datasets into actionable insights. Dedicated to applying mathematical theory to solve real-world data challenges, with a specific focus on data validation, process automation, and ensuring high-quality data standards in a professional analytics environment

SKILLS

- **Technical Languages:** **Python** (Advanced Data Manipulation), **SQL** (Joins, Window Functions, and CTEs), Beautiful Soup, yfinance.
- **Data Libraries:** Proficient in **Pandas** (Data Cleaning, Merging, and Grouping) and **NumPy** (Numerical Analysis and Multi-dimensional arrays),
- **Data Analysis & Visualization:** **Exploratory Data Analysis (EDA)**, **Data Validation**, and **Standardization**, developing proficiency in **Power BI**.
- **Quantitative Analysis:** **Statistical Modelling** and **Probability** (leveraging a strong M.Sc. Mathematics foundation).
- **Data Tools:** Microsoft Excel

ACADEMIC PROJECTS

Thesis: A Study on Topological Groups (M. Sc Project)

- Conducted a theoretical study of groups endowed with a topology such that Group operations are continuous.
- Analyzed properties of **Compact Groups** and **Homomorphisms**, strengthening foundations in Abstract Algebra and General Topology.
- Developed advanced **abstract reasoning** and **pattern recognition** skills applicable to high-dimensional data structures.

TECHNICAL PROJECTS

Financial Data Analytics & Visualization (IBM Certified)

- **Tools:** Python (Pandas, NumPy), BeautifulSoup, yfinance, Plotly.
- Developed an end-to-end pipeline to scrape and analyze 10+ years of historical revenue and stock data for Tesla and GameStop.
- Performed data wrangling and time-series normalization to identify market "regime shifts" and revenue-to-valuation correlations.

VIRTUAL INTERNSHIP PROGRAM

Quantum | Data Analytics Virtual Internship(via Forage) | February 2026

- Conducted deep-dive Customer Segmentation to identify high-value purchasing patterns for a retail client.
- Performed Data validation and cleaning on large-scale transaction data sets to ensure analytics accuracy.

CERTIFICATIONS

NPTEL Elite Certificate in Python for Data Science | IIT Madras | 2026

- **Result:** Elite (Score: 68%)
- **Key Skills:** Data Analysis (Pandas, NumPy), Visualization (Matplotlib, Seaborn), and Predictive Modeling.

IBM Data Science Professional Certificate | In Progress

- Specializing in: Predictive Modelling, Machine Learning with Python, SQL for Data Science and Data Visualization.

Databases and SQL for Data Science with Python — IBM (Via Coursera)

Skills: Advanced SQL (Joins, Subqueries), Database Management, Python-SQL Integration.

EDUCATION

[M.Sc. Mathematics](#) | CGPA :4.61/5.0 | University Rank 8

St Thomas College Palai,Kottayam | July 2025

- Graduated as a University Rank Holder.

[B.Sc. Mathematics](#) | CGPA: 9.60/10.0

Alphonsa College Palai,Kottayam | May 2023

- Awarded the National Scholarship for Postgraduate Studies based on academic excellence.

Class 12 | PERCENTAGE : 97.5

St Marys Higher Secondary School, Manarcadu